

Scribe Question 4: Inference and Usage of the Selected Bounds

Answer:

The probabilistic bounds derived in Milestone-2 provide both structural and risk-based insights into congestion behavior within the stochastic traffic model.

Inference from Jensen's Inequality:

Since congestion cost is modeled as a convex function $f(Q) = Q^2$, Jensen's inequality implies that variability in queue length increases expected congestion cost beyond the square of the mean. This directly motivates the use of a risk-aware objective function:

$$J = \mathbb{E}[Q] + \beta \text{Var}(Q),$$

where β controls the emphasis on congestion variability.

Thus, the system does not merely minimize average queue length but also penalizes fluctuations, preventing unstable or highly variable traffic states.

Inference from Markov's Inequality:

Markov's inequality provides an upper bound on the probability of excessive queue growth:

$$P(Q \geq Q_{\max}) \leq \frac{\mathbb{E}[Q]}{Q_{\max}}.$$

This spillover bound ensures that the probability of exceeding road capacity remains controlled. Similarly, for waiting time $W = Q/\mu$:

$$P(W \geq w) \leq \frac{\mathbb{E}[W]}{w}.$$

These bounds serve as conservative safety guarantees in the optimization framework. They allow the adaptive controller to maintain congestion risk within acceptable limits while selecting signal timings.

Overall Usage:

Together, the bounds:

- Justify the risk-aware objective formulation.

- Provide safety constraints against spillover.
- Support proactive traffic regulation under stochastic demand.

They integrate directly into the stochastic optimization problem governing signal control.

Scribe Question 5: Cross-Milestone Consistency and Change

Answer:

In Milestone-1, uncertainty was identified qualitatively through key random variables such as vehicle arrivals, queue length, saturation flow, and vehicle speed. The system was described conceptually as stochastic and decision-making was modeled using a Las Vegas randomized strategy.

In Milestone-2, we formalized the probabilistic framework mathematically:

- Arrivals were modeled as time-varying Poisson random variables $A_i(t) \sim \text{Poisson}(\lambda_i(t))$.
- A multivariate arrival vector $A(t) = (A_1(t), A_2(t), A_3(t), A_4(t))$ was introduced for the four-way junction.
- Queue dynamics were explicitly modeled:

$$Q_i(t+1) = \max\{Q_i(t) + A_i(t) - D_i(t), 0\}.$$

- Randomized scheduling defined direction selection $S(t)$ based on queue proportions.
- The system state $X(t) = (Q(t), S(t))$ was shown to form a controlled Markov process.
- A risk-aware optimization objective $J = \mathbb{E}[Q] + \beta \text{Var}(Q)$ was formulated.

Thus, the project evolved from qualitative uncertainty modeling to a rigorous stochastic control framework incorporating system dynamics, probabilistic bounds, and optimization constraints.

Scribe Question 6: Open Issues and Responsibility Attribution

Answer:

Although Milestone-2 establishes a formal stochastic control model, several probabilistic aspects remain open:

1. **Arrival Independence Assumption:** Independence across directions and Poisson structure may not fully capture correlated urban traffic patterns.

2. **Service Rate Modeling:** Departure rate μ_i is treated as constant, whereas real traffic exhibits variability due to driver behavior and road conditions.
3. **Markov Property Validation:** The system is modeled as a controlled Markov process, assuming next-state dependence only on current state. Empirical validation is required.
4. **Tightness of Risk Bounds:** Markov inequality provides conservative bounds. Tighter concentration inequalities may improve safety estimation.
5. **Parameter Selection:** The choice of risk-weight parameter β in the objective function requires calibration.

Responsibility Attribution:

- In Data/Simulation: Validate assumptions made for distribution and independence structure.
- In Modeling: Refine service rate and queue evolution modeling.
- For Probability Analysis: Explore tighter concentration bounds.
- For Optimization: Calibrate and evaluate the risk parameter β .

Addressing these issues will strengthen the alignment between the theoretical stochastic model and real-world adaptive traffic behavior.

End of Submission